Orthographic projection

Higher · Drawing & spatial awareness · Answers

Answers and marking guidance

Award one mark per correct, developed point unless stated. Other valid answers should be credited.

2 Calculate the length of the slot.

- 200 − 62 − 72 = 66

No marks for: Answer without units is fine; working not required

3 Quick check

- a) 13
- b) Thin dashed lines
- c) At 90° to the cut surface
- d) Show the true shape of an inclined surface

4 Key terms

- 1 – A
- 2 – F
- 3 – D
- 4 – E
- 5 – C
- 6 – B

5 Calculate the distance from the datum to (i) the centre of the external radius and (ii) the centre of the internal fillet.

- (i) 15 − 2.5 = 12.5 (the centre of an external radius sits inside the material)
- (ii) 4 + 1.5 = 5.5 (the centre of an internal fillet sits in the open space)

No marks for: Working is not required

6 Explain why an auxiliary view of this face would be included on the production drawing.

- It shows the true shape of the inclined face (cause: projected at 90° to it)
- So the holes appear as true circles and can be dimensioned accurately (effect)
- In normal views the face is foreshortened and the holes appear as ellipses